FEMA Fact Sheet

PTS Innovations Consequence Solution: How to Download and Launch the Prototype Marimo Notebook

Consequence modeling plays a vital role in actuarial assessments, scenario evaluations, and Benefit Cost Analyses (BCAs). Reliable Annualized Average Loss (AAL) estimates help set insurance premiums, manage financial risk, support emergency planning, and justify investments in flood mitigation projects.

To support these needs, the PTS Innovations Account developed a **prototype Consequences Solution** through the Standard Operations 3.1 Project, titled Inland Consequences. This project created a flexible, modular framework designed to quickly calculate AAL data. The core of the solution is a Python based library that incorporates the latest inland consequence research and modernizes the existing FEMA Coastal Loss Tool by migrating it into Python.

This **prototype notebook** demonstrates how to access and use the library through a Marimo notebook environment. Advanced users can expand this notebook or develop their own workflows using the Python library. This guide explains how to download the Inland Consequences repository from the FEMA FFRD GitHub site and provides step by step instructions for download and launch the Prototype Notebook.

For more details on the methodology and technical implementation, please refer to the Consequence Solution documentation site.

Prerequisites:

This tool requires the ultra-violet (uv) package manager. If you don't already have uv installed, follow the steps below.

1) Open Windows PowerShell.

2) Copy and paste the command below, then press Enter:

```
powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

Note: If you do not have administrator privileges, you may need to contact your IT team to install uv for you.

For the best experience, we recommend running this notebook in **Visual Studio Code (VS Code)**, although other software may also work. The instructions below assume that **VS Code** is already installed on your computer.

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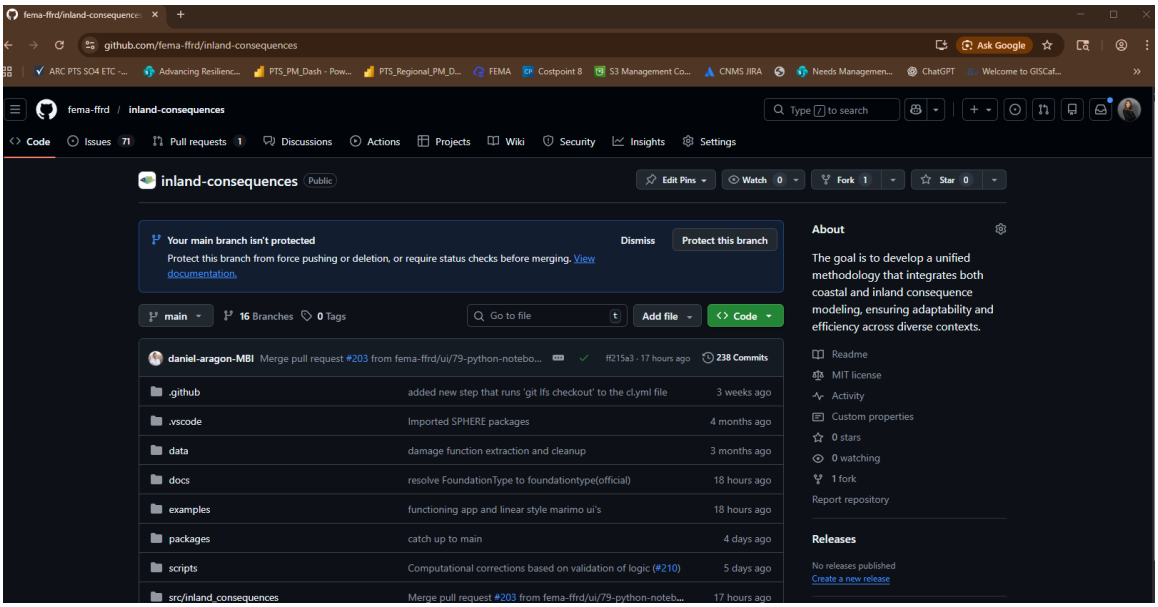
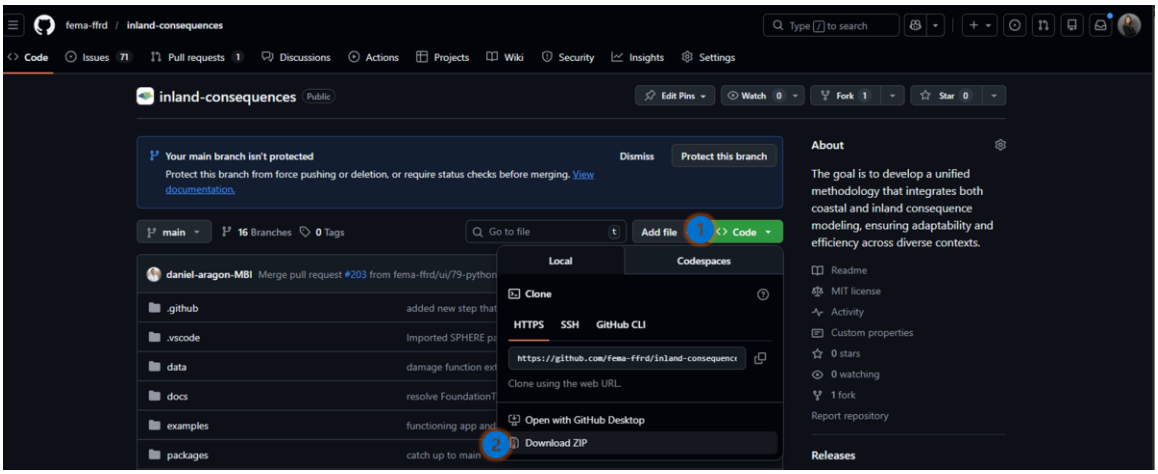
Download and Open the Prototype Miramo Notebook

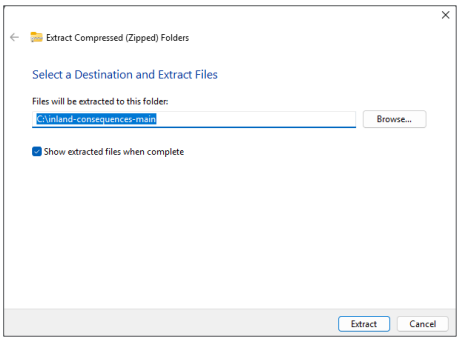
1. Open your web browser and go to:
   https://github.com/fema-ffrd/inland-consequences



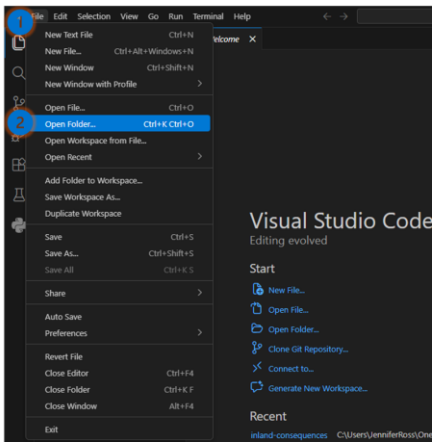
2. On the page, click the green **<> Code** button, then select **Download ZIP**.
   Save the ZIP file to a location on your computer where you can easily find it later.

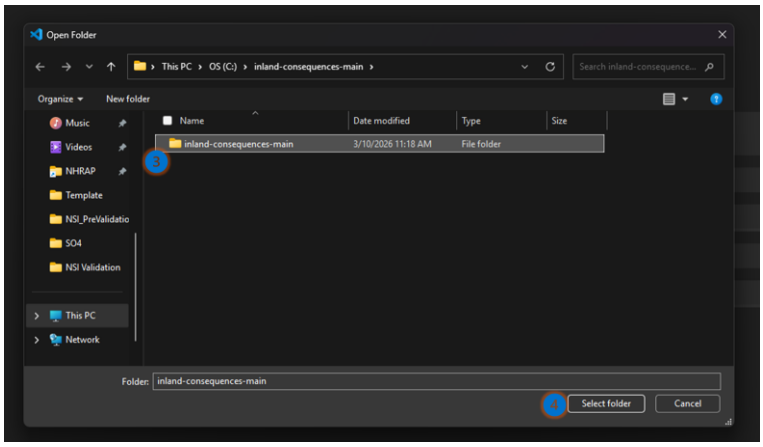


3. Extract (unzip) the downloaded ZIP file.

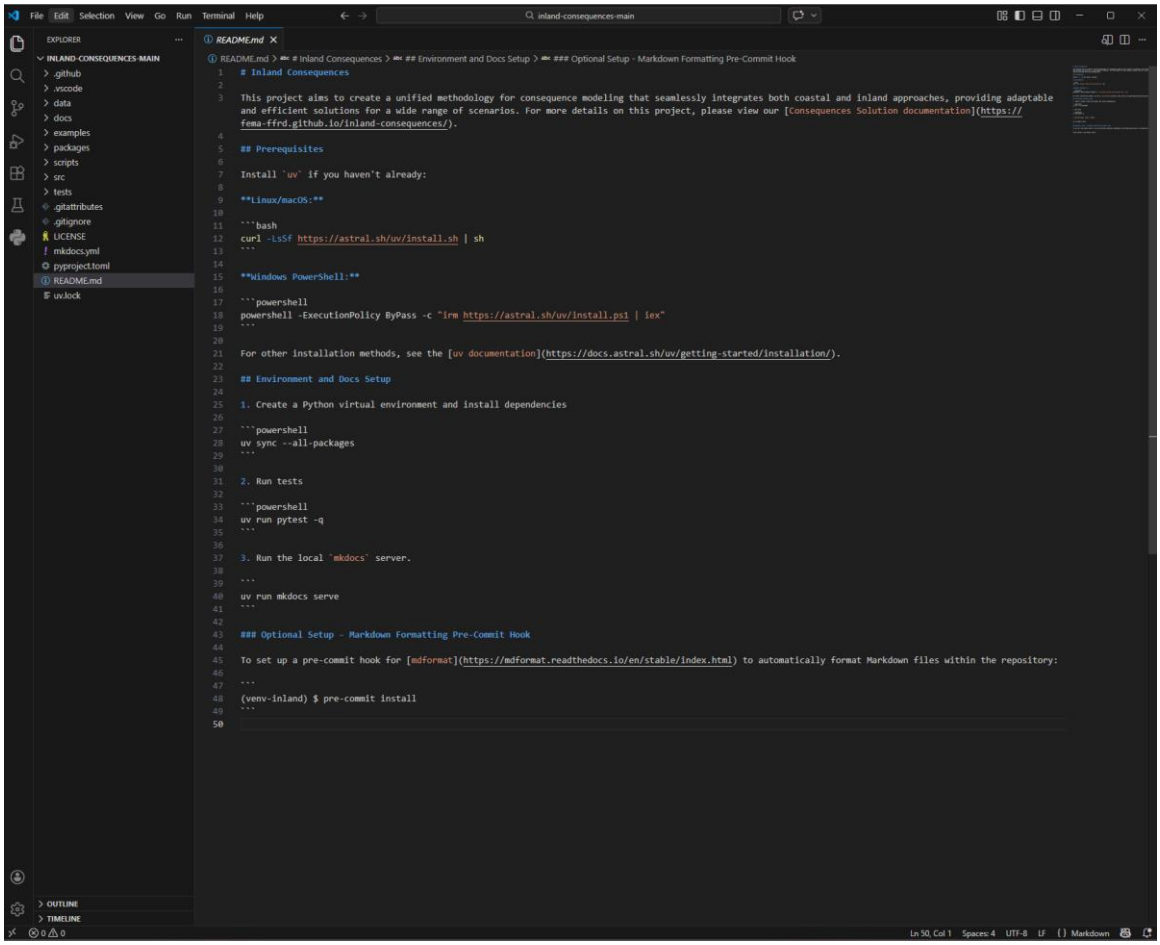


4. Open **Visual Studio Code (VS Code)**. In the top ribbon, select **File** > **Open Folder**. Browse to the folder you just unzipped, select it, and click **Open Folder**.

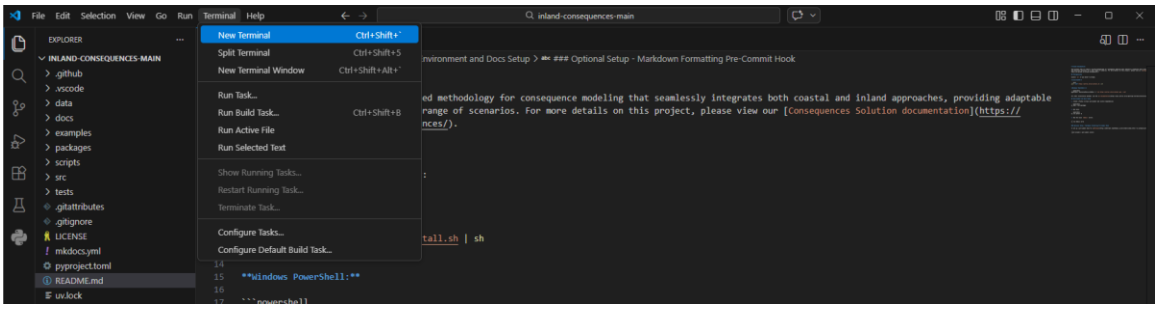




5. In the left-hand **Explorer** panel, you should now see the **inland-consequences** folder. Select **README.md** to open the README file.



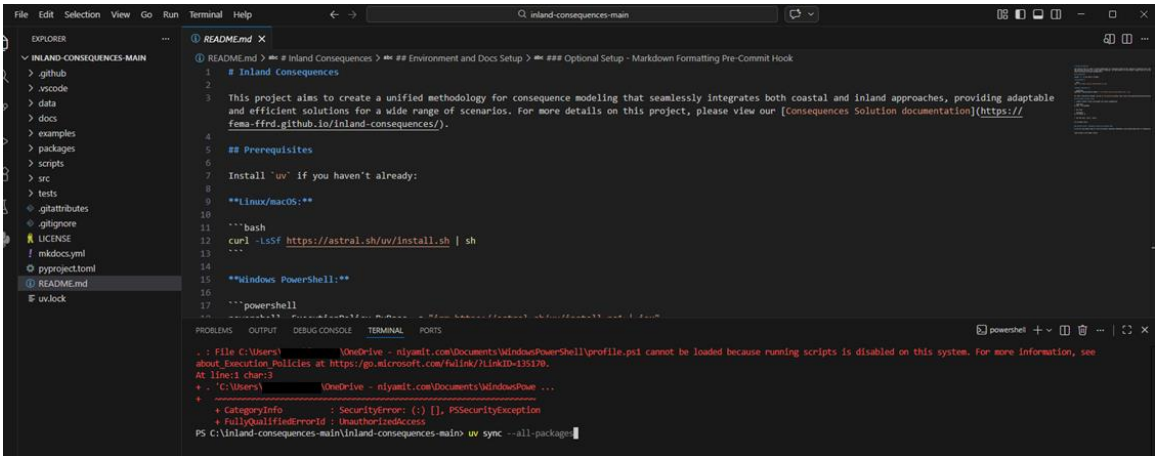
6. If the terminal is not already open, go to the top menu and select **Terminal** > **New Terminal**.



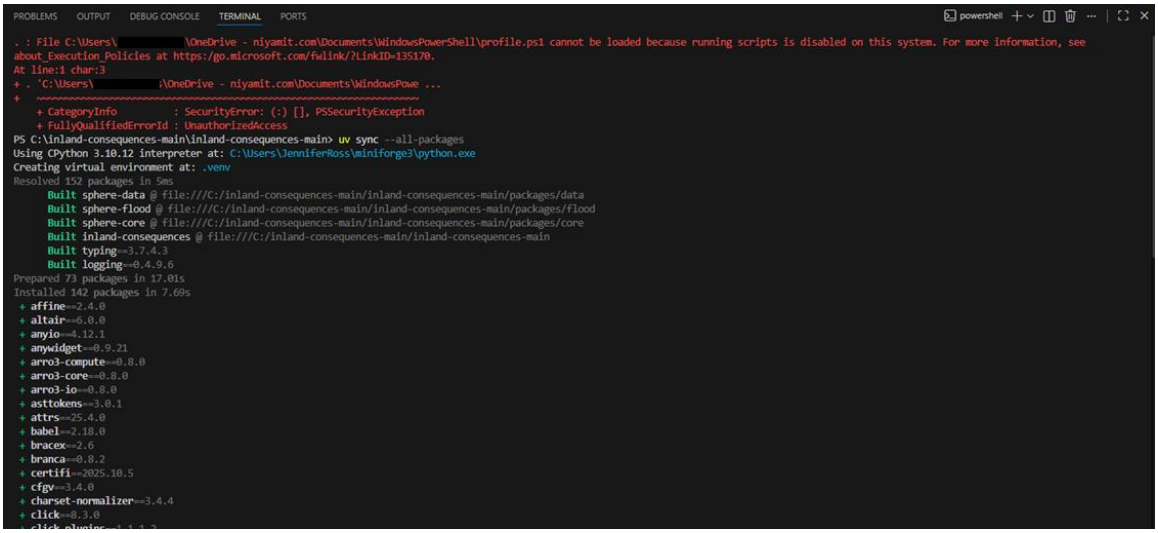
7. In the terminal, copy and paste the following command, then press **Enter**:

Code:

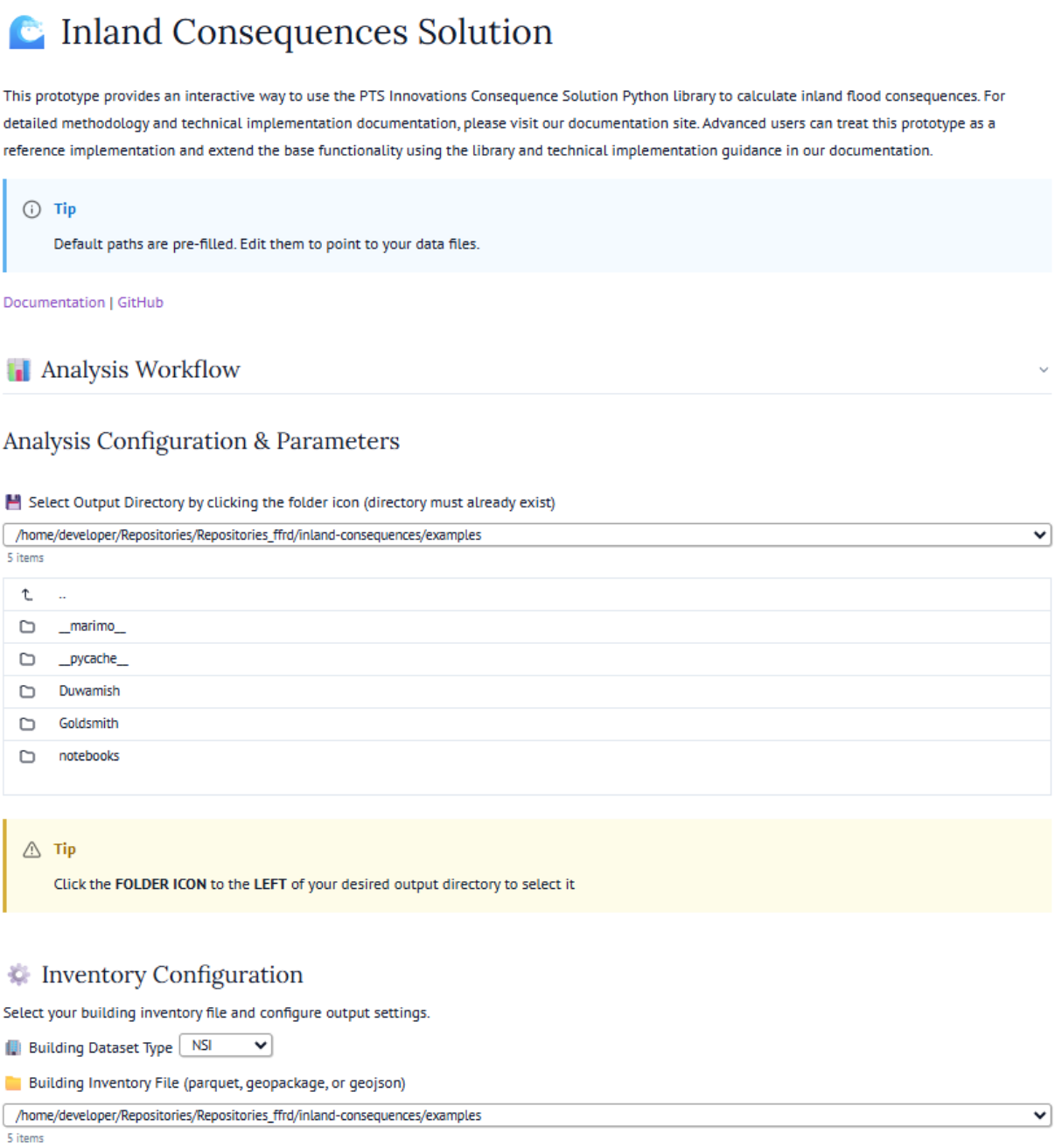
```
uv sync --all-packages
```



8. This command installs the packages required to run the tool.



9. After the installation is complete, copy and paste the following command into the terminal, then press **Enter**:

   Code:
   ```
   uv run marimo run examples/notebooks/run_inland_consequences.py
   ```

10. After you run the command, the prototype notebook should open automatically in your default web browser from your local machine. If a browser window does not open, copy the URL shown in the terminal and paste it into a new browser window.